

Momentum: A Seer on Player Performance and Game Swings Based on Machine Learning

Abstract

The 20-year-old Carlos Alcaraz's tussle with veteran Novak Djokovic in the men's singles final at Wimbledon 2023 was a thrilling one. Carlos Alcaraz's victory was described as "the crowning of a new king". While this ignited our passion for the game, it also prompted us to think about the factors that contribute to winning and the match situation. In order to solve this problem and unveil the mystery of "momentum", we build Player Performance and Match Dynamics Analysis Model and Match Fluctuations Prediction Model based on **LM-BP Neural Network** and **Support Vector Machine (SVM)**, respectively, to carry out an in-depth study on this topic.

Before modeling, we performed the necessary data processing, in order of the initial classification of indicators, data cleaning, correlation analysis, and data normalization.

For Problem 1-2, we first set up a player performance assessment index system, using existing data to reflect their serving and returning abilities, and using the normalized values of both as momentum, which serves as a quantitative indicator of the performance of players. Then, we optimized the traditional BP neural network using the Levenberg-Marquardt algorithm to calculate and analyze players' momentum based on machine learning of historical data. We found that the trend of momentum is highly similar to the trend of players' scoring rate, which strongly confirms the validity of the quantitative results of players' performance. Then, we used **Monte Carlo method** to establish the initial value of a random game, and after iterating the LM-BP neural network, we finally visualized the game flow in the form of a network chain diagram. Based on the Spearman correlation test, we found a strong correlation between momentum and the difference in player scores per game, concluding that momentum plays an important role in the game.

For Problem 3-4, we use the change rate of the difference in scores between the two players in each game as the indicator to determine the fluctuation of the game. We use SVM model to predict the fluctuation of the match and determine the judgment interval and corresponding indicator values for the fluctuation change. Then, we used the MRMR algorithm to calculate the importance scores of the characteristic variables, and found that unforced errors, number of breaks of serve, and number of aces had the greatest impact on the match fluctuations, and the importance scores of each of them were 0.224, 0.175, and 0.168, respectively. Based on the above conclusions, we gave suggestions to the players to deal with the new opponents. We used the final match between Carlos Alcaraz and Novak Djokovic as an example to predict the match fluctuation with an accuracy of 83.673%. Then, we shared ideas for future model improvements and made a qualitative evaluation of the model's generalization ability.

Finally, we performed a sensitivity analysis of the model, summarized its strengths and weaknesses, and wrote a data report.

Key words: Momentum; LM-BP neural network; SVM; Monte Carlo method

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1. Introduction

1.1 Background

Carlos Alcaraz finally beat Novak Djokovic to claim the title in the much-anticipated Gentlemen's singles final at Wimbledon 2023 after a tense and scorching five-set match. It was a crushing blow for Djokovic. What made this match so tumultuous and heart-wrenching? Some believe it had to do with "momentum", and experience has shown us that the side that seems to be dominating can sometimes have spectacular swings of multiple points or even games. Exploring the relationship between game swings and success is the key to unraveling the mysterious power of momentum. Paying close attention to a player's on-court performance when he scores (or loses) points, grasping the dynamics of the match, predicting match fluctuations, and analyzing the factors that influence them are all important guides to further training and development, as well as adjustments in match strategy. We have studied the Wimbledon Gentlemen's singles match rules and mapped out the flowchart of the scoring rules as follows.

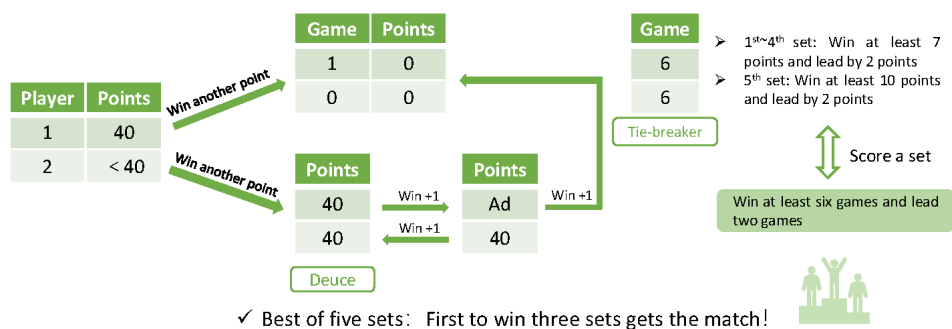


Figure 1 Wimbledon Gentlemen's Singles Match Scoring Rules Flowchart

1.2 Restatement of Problem

Considering the background information given in the question and the data provided in the appendix, we need to solve the following problems:

- **Task 1:** Build a model that can capture and visualize the flow of the match when the score is updated and apply it to one or more matches. In addition, our model should quantify how good or poor a player's performance is, identifying which player is performing better and to what extent at a particular time in the match.
- **Task 2:** Using the established model to study the relationship between player performance fluctuations and winning streak, and explore the role of "momentum" in the competition.
- **Task 3:** Use the match data given in the appendix to develop a model to predict fluctuations in the match and analyze the indicators that may affect the fluctuations.
- **Task 4:** Provide players with advice on how to deal with new players, taking into account differences in "momentum" fluctuations in past matches.

- **Task 5:** Test our model in other types of matches. Check the accuracy of the model's predictions and evaluate the model's generalization capability to other matches. Based on this, explore future improvements to the model.

1.3 Literature Review

Research on the influencing factors of tennis matches usually uses methods such as regression analysis, discriminant analysis, and factor analysis. Jiang Ting and Li Qing^[1] used a logistic regression model to explore the effects of three variables: player performance, personal traits, and competition environment on victory or defeat. Their research showed that the type of match does not have a significant impact on victory or defeat, while serving, receiving, and breaking point scoring rates play an important role. In addition to the scoring rate of first serve being the best predictor of winners and losers, their study also concluded that player rankings were effective in predicting the outcomes.

Tennis match results prediction has become a research hotspot in recent years, with the most common being prediction models represented by Markov chains. However, traditional Markov prediction models based on the hierarchical structure of tennis scoring systems only use the probability of each player's serve score to calculate their probability of winning the match, thus ignoring various influencing factors, resulting in low prediction accuracy. On this basis, Zhang Rong^[2] optimized the Markov chain model by incorporating two variables: the intrinsic difference between player serving and receiving, and the impact of venue type into the model, improving the accuracy of model prediction and proposing possible improvements in tracking player momentum. Stephanie Kovalchik and Machar Reid^[3] proposed a dynamic in-match prediction method that by combining a Pre-Match calibration method with a dynamic empirical Bayesian updating rule, thereby incorporating both useful pre-match information and relevant in-match information in order to update pre-match expectations as the match progresses.

1.4 Our Work

Figure 2 shows the process of our work:

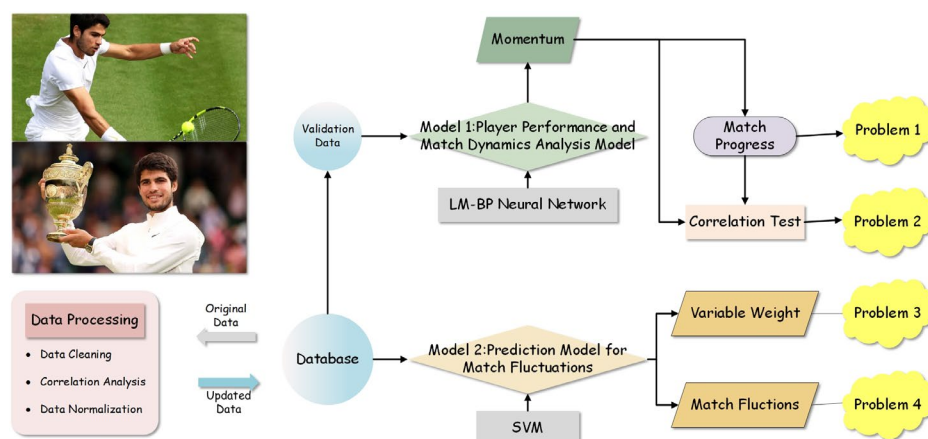


Figure 2 Our Work Flowchart

2. Assumptions and Symbols

2.1 Assumptions and Justifications

In this section, we introduce the following global assumptions so as to simplify our model; assumptions specific to each model will be stated and justified in their corresponding model introduction and setup.

- **Assumption 1 The rules of the tennis matches are all based on those of the Gentlemen's singles at Wimbledon, with a best-of-five system.**

There are some differences in the rules of the Four Grand Slams of tennis. The data provided in the appendix is from the results of the Wimbledon Gentlemen's Singles Match and is the data support source for our model. So for simplicity, when discussing tennis matches, we default to using the rules of Wimbledon Gentlemen's singles.

- **Assumption 2 This paper does not consider the outcome of doubles matches, so there are no complex team relationships.**

When considering tennis as an individual sport, there is no time limit on a tennis match until one of the players wins and there is no possibility of a tie.

- **Assumption 3 The weather is suitable and stable for a tennis match.**

The Wimbledon Open is played outdoors on grass, and when it rains, it's windy and it's hot, it has a strong inhibiting effect on a tennis player's performance.

2.2 Symbols and Definitions

Before we analyze the problems, it is necessary to clarify the symbols that we will use in our discussion. These are shown below in Table 1:

Table 1 Symbols and Definitions

Symbols	Definitions
a_1	each player's offensive situation
a_2	each player's defensive situation
a_3	each player's error situation
a_4	each player's physical condition
a_5	other factors
x_1	serve ability
x_2	return ability
η	the learning rate
l	the number of hidden layers
n	the number of input layer nodes
m	the number of output layer nodes

3. Data Processing

Referring to the existing studies such as P. Gorgi and S. J. Koopman^[4], we can know that all the men's match data after the first two rounds of 2023 Wimbledon Open provided in the appendix can be mainly used to characterize each player's offensive

situation a_1 , defensive situation a_2 , error situation a_3 , physical condition a_4 and others a_5 (as shown in Figure 3). In order to enhance the expressive ability of the features and improve the efficiency and accuracy of the model, before building the model, we perform data cleaning, correlation analysis and data normalization on the raw data to get a complete and accurate set of data.

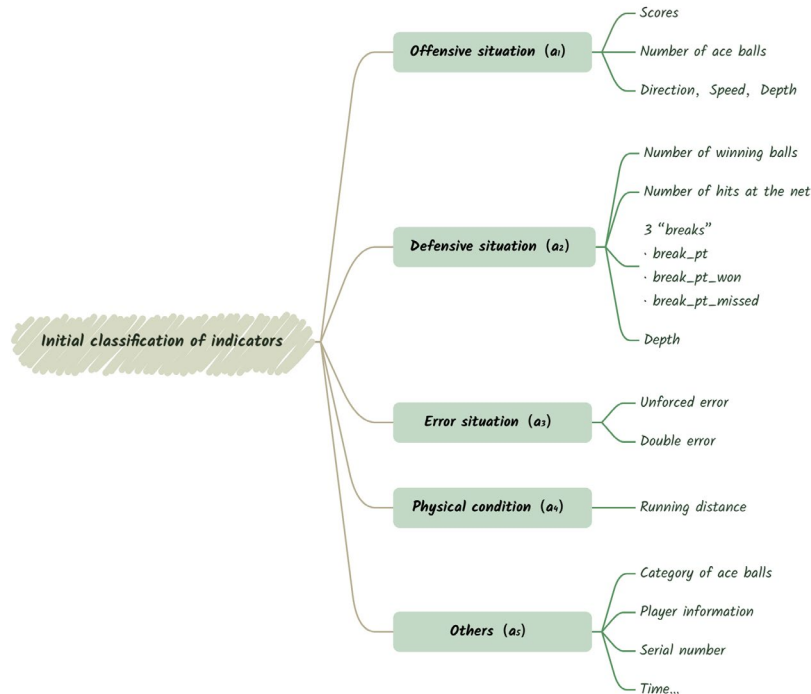


Figure 3 Tree diagram of initial classification of indicators

3.1 Data Cleaning

In compiling the data, based on the serve speed scatterplot, we found that of the 7258 games, Daniel Elahi Galan vs. Mikael Ymer, Guido Pella vs. Roman Safiullin, and Denis Shapovalov vs. Liam Broady, for a total of 3 matches, all measured the serve speed of NA, as shown in Figure 4. According to the rules of tennis, it is known that when a server misses a serve, the speed of serve will be measured as NA, but the probability of occurrence is low at 4.86%. Therefore, we remove and ignore the outliers of these 488 games in order to better characterize the tennis players' on-court performance.

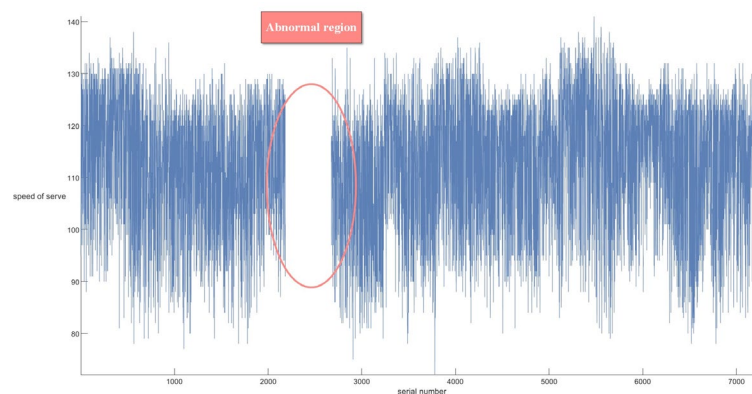


Figure 4 Speed of serve scatterplot

3.2 Correlation Analysis

(1) Correlation analysis for the category of ace balls

In tennis, a superb serve, a fast return, or an opponent's miscue can result in the opponent not touching the ball and allowing us to win the point, which we call an ace. A beautiful ace requires a combination of speed, drop shot and tactics, and also depends largely on the player's ability to receive and serve the ball. The ace ball is categorized into forehand and backhand. We calculated the hit rate of each type of ace ball by counting the number of balls and hit balls of each ace type (as shown in the table below). It can be found that the hit rates of forehand and backhand ace balls are very close.

Table 2 The scoring rate of different ace balls

Category of ace balls	No. of balls	number of goals scored	goal-scoring rate
Forehand	459	255	0.55
Backhand	1072	542	0.51

In addition, there is a highly linear relationship between the category of ace balls and the number of ace balls (e.g., Figure 5), so we did not use it as a factor to assess player performance to avoid the multiple covariance problem.

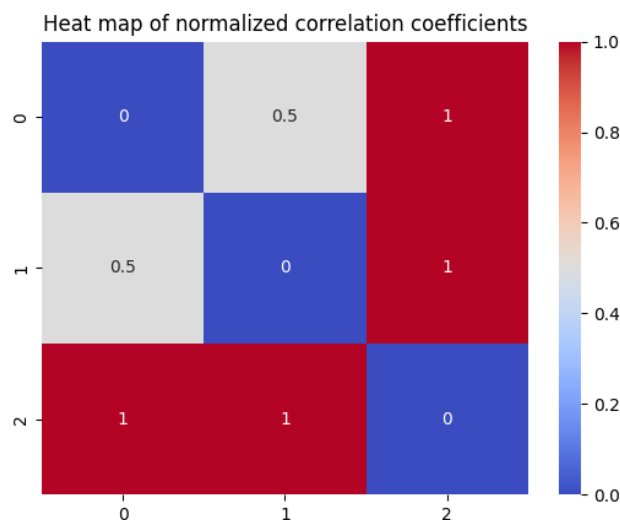


Figure 5 Heat map of correlation coefficients between ace ball count and ace ball category

- 0 and 1 indicate the number of ace balls hit by both players respectively, and 2 indicates the category of ace balls.

(2) Correlation analysis for direction of serve and depth of serve (or return)

The appendix gives a total of 9 labels for serve direction and depth of serve (or return), and we found that different serve directions and depths of serve (or return) had little effect on the scoring rate by creating frequency histograms and line graphs of scoring rate. We analyzed the reasons for this and concluded that the players' choices or control of serve direction and depth of serve (or return) are more dependent on their personal habits and common sense of the tennis match, and therefore have almost no impact on the winners and losers.

- ♦ The term "frequency" refers to the number of times a characteristic occurs in a variable.

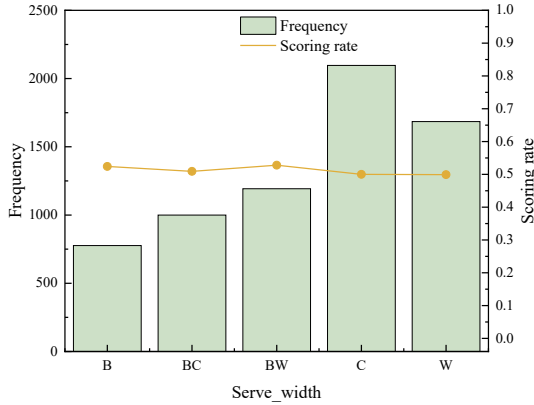


Figure 6 Direction of Serve Frequency Histogram - Scoring Rate Line Graph

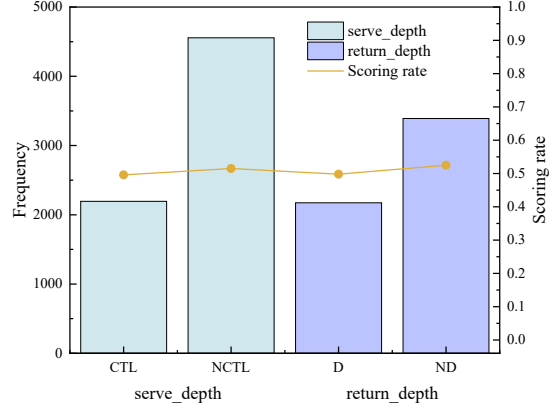


Figure 7 Depth of Serve (or Return) Frequency Histogram - Scoring Rate Line Plot

3.3 Data Normalization

It is easy to know by checking the indicators that there is a difference in magnitude between serve speed and running distance and other indicators. Before the model is built, we need to make sure that all the data are in the same magnitude, so we use Min-Max normalization method to process the data, so that the values of these two indicators are compressed to [0,1]. We used the following formula to normalize the data:

$$y = \frac{t - t_{min}}{t_{max} - t_{min}} \quad (1)$$

These analyses provide the basis for the subsequent model establishment.

4. Model 1: Player Performance and Match Dynamics

Analysis Model Based on LM-BP Neural Network

In this section, we build a model to evaluate player performance and track it dynamically in relation to match conditions, and finally use our model to visualize the flow of the match. Player performance focuses on serve ability x_1 and return ability x_2 . Momentum is closely related to a player's performance, and we believe that momentum can be used to some extent as an indicator to quantify how well a player is performing. Based on the preliminary analysis we have done in data processing, we first establish the evaluation indicator system of player performance, and then build a high-precision LM-BP neural network analysis model using historical data, so that we can calculate the players' momentum directly from the measurement indicators, so as to evaluate and compare the players' performances with the progress of the match, and make a dynamic analysis of the match conditions, and then draw a visualization graph of the course of the match.

4.1 Establishment of a System of Indicators for Assessing Player Performance

By checking the official website of the Association of Tennis Professionals (ATP) and the websites of related sports events^[5-6], it can be seen that experts generally believe that these technical indicators, i.e., the success rate of the first serve, the ace balls, the double error, the unforced error, the second serve scoring rate, the reception-serve scoring rate, and the total scoring rate, can reflect the performance ability of both sides of the match, and are easy to be judged and recorded in the statistics. Huang^[7] studied the role of the above indicators on the winning and losing results of professional gentlemen's tennis singles matches, and the prediction accuracy was 94%. In summary, we believe that the above indicators provide a relatively comprehensive assessment of a player's performance.

On the basis of the global data processing in section 3, we sieve and professionally convert the indicator data on a player basis. The serve ability x_1 is characterized by the first serve hitting rate, average ace balls per game, double error probability, points at the net, number of hits at the net, and serve scoring rate. The return ability x_2 is characterized by winning balls, average returning balls, return scoring rate, points at the net, number of serves at net, and unforced errors. Then, we analyze the performance of the players by combining serve ability x_1 and return ability x_2 .

4.2 Calculation and Analysis of Player Momentum

BP neural networks are good at finding patterns from input and output signals, and have strong non-mapping ability^[8]. Due to their powerful data recognition and simulation ability, they have been widely used in the fields of pattern recognition, image processing, fault diagnosis, financial forecasting and market analysis in recent years. However, the traditional BP neural network embodies the problems of slow error convergence and easy to fall into local minima in practical applications. In order to overcome the limitations of BP neural networks, scholars at home and abroad have done a lot of research on the optimization of BP neural network algorithm, and in the study of BP neural networks based on the defects of improvement algorithms, there are mainly additional momentum method, Adaptive learning rate algorithm, the quasi-Newton method, the LM algorithm and so on.

Due to the complexity of player performance indicators and the large amount of match sample data, it is more suitable to optimize the BP neural network from the perspective of numerical optimization. The LM-BP neural network inherits the structure, connection method, and error transmission method of traditional BP neural network. On this basis, it replaces the single gradient descent method of traditional BP neural network and uses approximate second-order reciprocal information to balance local and global when searching for the optimal solution^[9], avoiding the problem of neural network getting stuck in local minima when solving, and effectively accelerating the learning speed of the network, making it more suitable for processing large amounts

of data. In view of the advantages of the LM-BP neural network mentioned above, we choose to use it to calculate and analyze players' momentum.

4.2.1 Training Process of BP Neural Network

BP neural network is a multi-layer feed-forward neural network composed of multiple artificial neuron models, including input layer, hidden layer and output layer, which is characterized by the forward transmission of information and reverse transmission of error, and the adjustment of its connection weights is to use the error between the actual output and the desired output to correct the connection weights of each layer of the network layer by layer, from the backward to the forward. The topology of the BP neural network and the artificial neuron models are shown in Figure 8.

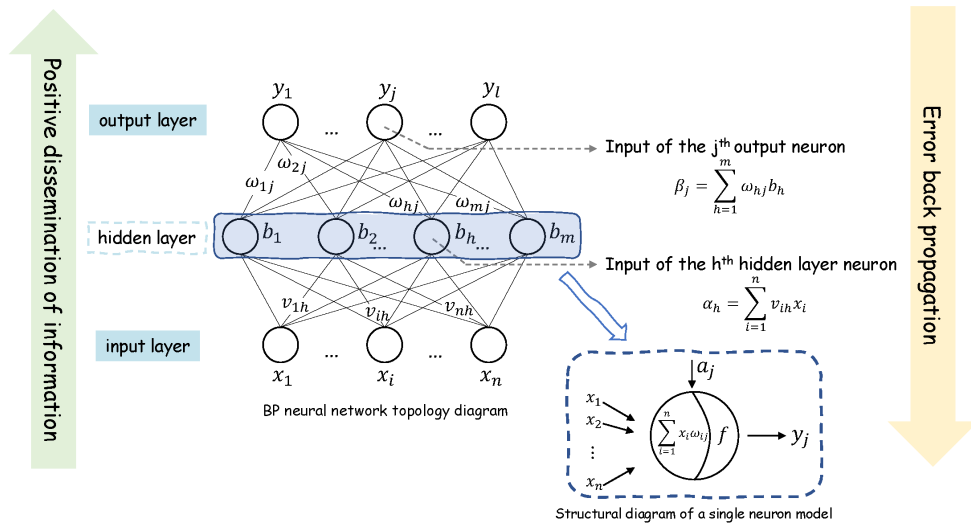


Figure 8 Topology of BP Neural Networks and Hidden Layer Artificial Neuron Model

The specific BP neural network building process is shown below:

Step 1: Network Initialization

Based on the input $X = (x_1, x_2, \dots, x_n)$ and the desired output $Y = (y_1, y_2, \dots, y_n)$, try multiple times to obtain the number of neurons in the input layer, hidden layer, and output layer with the best prediction accuracy. Initialize the connection weights and thresholds between neurons in each layer, and provide the learning rate η and activation function sigma function. The formula for the activation function sigma is as follows:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

Step 2: Hidden Layer Output Calculation

Based on the input vector X , the connection weights ω_{ij} of the input and implied layers and the implied layer threshold x_i , the implied layer output Y is computed:

$$y_j = f(\sum_{i=1}^n \omega_{ij} x_i - a_j), \quad j=1, 2, \dots, m \quad (3)$$

- ◆ Where m is the number of nodes in the hidden layer

Step 3: Output Layer Output Calculation

Based on the output vector Y of the hidden layer, the connection weights ω_{jk} , and the threshold b_k to calculate the actual output O of the BP neural network:

$$O_k = f(\sum_{j=1}^m \omega_{jk} y_j - b_k), \quad k = 1, 2, \dots, l \quad (4)$$

- Where l is the number of nodes in the output layer

Step 4: Error Calculation

Calculate the overall error E based on the actual output O and the desired output Y :

$$E = \frac{1}{2} (D - O)^2 = \frac{1}{2} \sum_{k=1}^l (d_k - o_k)^2 \quad (5)$$

Step 5: Weight Update

If the error does not meet the accuracy requirements, the weights ω_{ij} and ω_{jk} should be updated according to the following formula:

$$\omega_{ij} = \omega_{ij} + \eta (\sum_{k=1}^l \delta_k^o \omega_{jk}) y_j (1 - y_j) x_i, \quad j = 1, 2, \dots, m, k = 1, 2, \dots, l \quad (6)$$

$$\omega_{jk} = \omega_{jk} + \eta \delta_k^o y_j, \quad j = 1, 2, \dots, m, k = 1, 2, \dots, l \quad (7)$$

- Where $\delta_k^o = (d_k - o_k) o_k (1 - o_k)$, η is the learning rate

Step 6: Determine If the Algorithm Iteration Is Over

Set the accuracy requirement in advance. When the overall error E reaches the accuracy requirement, the iteration ends; if not, return to Step 2.

According to the above steps, the flowchart of BP neural network training is shown in Figure 9:

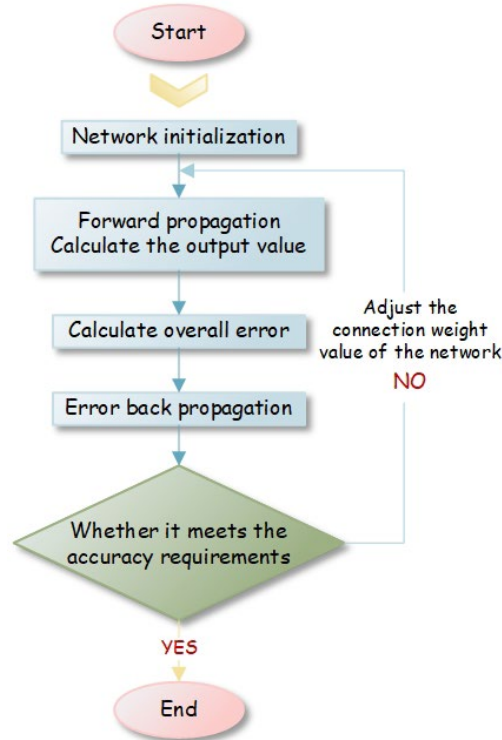


Figure 9 BP neural network training flowchart

4.2.2 Optimization of BP Neural Networks Based on Levenberg-

Marquardt Algorithm

Although BP neural network has powerful data analysis and data simulation ability, it has overfitting phenomenon in predicting players' momentum, and the gradient descent algorithm also makes it fall into local optimal solution. In order to make the prediction results more accurate, we choose the LM algorithm to change the original weight updating method from the perspective of numerical optimization to inhibit the network from falling into local minima.

The LM algorithm is derived from Newton's method and gradient descent^[10]. In order to apply Newton's algorithm to the case where the Hessian matrix is non-positive definite, this algorithm tries to avoid computing the Hessian matrix by computing the positive definite matrix. Meanwhile, the LM algorithm introduces a scaling factor μ to control the size of weight adjustment, which effectively accelerates the learning speed of the network. Meanwhile, the two-dimensional search method is also very effective in suppressing the network from falling into local minima.

To better demonstrate how the LM algorithm works in BP neural networks, we utilize pseudo-code to illustrate the algorithmic structure and principles of the LM algorithm, as shown below:

Algorithm 1: Levenberg-Marquardt method

Input: Historical indicator data for tennis players
Output: Momentum value for tennis players

```

1:  begin
2:     $k := 0; v := 2; x := x_0$ 
3:     $B := J(x)^T J(x); g = J(x)^T f(x)$ 
4:     $found := (\|g\|_\infty < \varepsilon_1) ; \mu := \tau \cdot \max \{a_{ii}\}$ 
5:    while (not found) and ( $k < k_{max}$ ) do
6:       $k := k + 1; Solve(B + \mu I)h = -g$ 
7:      if  $\|h\| < \varepsilon_2(\|x\| + \varepsilon_2)$  then
8:         $found := \mathbf{true}$ 
9:      else
10:        $x_{new} := x + h$ 
11:        $e := (F(x) - f(x + h)) / (L(0) - L(h))$ 
12:       if  $e > 0$  (step acceptable) then
13:          $x := x_{new}$ 
14:          $B := J(x)^T J(x); g = J(x)^T f(x)$ 
15:          $found := (\|g\|_\infty < \varepsilon_1)$ 
16:          $\mu := \mu \cdot \max \{1/3, 1 - (2e - 1)^3\}; v := 2$ 
17:       else
18:          $\mu := \mu \cdot v; v := 2 \cdot v$ 
19:       else if
20:       else if

```

21: end while

In the above pseudo-code, $f(x)$ is the residual function obtained after one training of the neural network, and the expression of $F(x)$ is $F(x) = \frac{1}{2} \sum_{i=1}^N \|f(x)\|^2 = \frac{1}{2} \|f(x)\|^2 = \frac{1}{2} f(x)^T f(x)$, J is the Hessian matrix of $f(x)$, and g represents the gradient.

4.2.3 Model Training Effect Detection

Before building the LM-BP neural network, we organize the data of the relevant indicators characterizing the player's serve ability x_1 and return ability x_2 as an array of input layers, and take the serve scoring rate and for the return scoring rate as the desired output, so as to determine the number of neurons in the input and output layers.

During the training process, in order to fully utilize each data and prevent overfitting, we randomly divided the data into training, validation, and testing sets in a ratio of 70:15:15, and set the hidden layer parameter to 9. We measure the network performance by the mean square error between the actual output and the desired output, and the iteration stops when the number of training times reaches 1,000.

For the sake of meeting the accuracy requirements, we build LM-BP neural networks for serve ability x_1 and return ability x_2 respectively for multiple long-term predictions, and get the LM-BP neural network with the best fitting goodness, as shown in Figure 10. The R values of the training set, validation set, test set and the overall data in serve ability x_1 and return ability x_2 are mostly greater than 0.95, with an average of 0.976, indicating that our model is well trained and can effectively reflect the performances of players.



Figure 10 Model training effect goodness-of-fit analysis plot

4.3 Model Application 1: Quantitative Expression and Dynamic Monitoring of Players' Performance

In tennis matches, the probability of the server scoring is usually much higher, and we take this into account. Now, we take the match between Carlos Alcaraz and Novak Djokovic as an example:

We use $\frac{x_1}{x_1+x_2}$ as the momentum of the serving side and $\frac{x_2}{x_1+x_2}$ as the momentum

of the other side, which will continue to change as the game progresses. Next, we calculate the scoring rates of Carlos Alcaraz and Novak Djokovic, and analyze the relationship between scoring rates and momentum, as shown in Figure X. The curve graph shows that the scoring rate and momentum are highly similar in trend and values, which not only confirms the rationality of the quantitative results of the player's performance, but also reflects the fluctuation of the player's performance with the progress of the game.

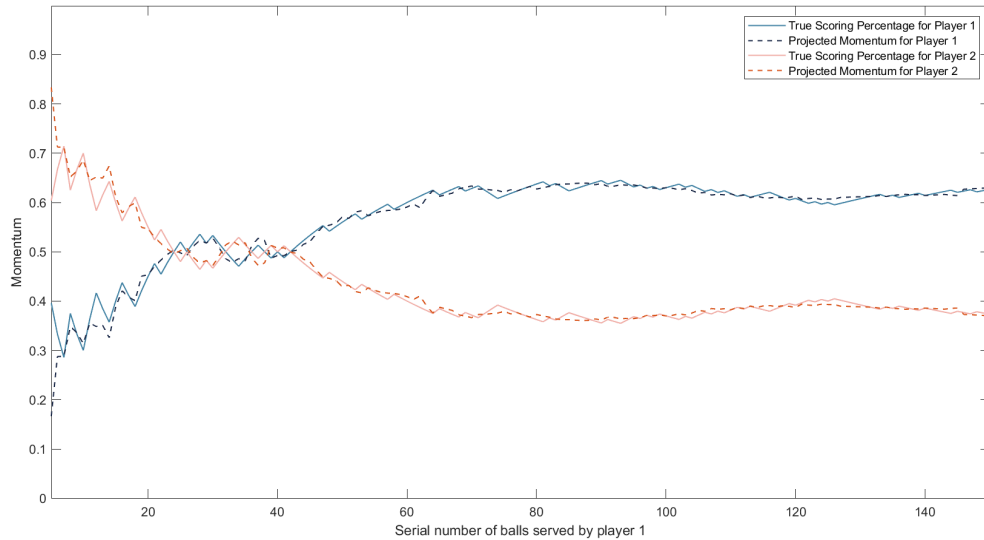


Figure 11 Trend charts of potential energy and scoring rate of Carlos Alcaraz vs. Novak Djokovic

4.4 Model Application 2: Visualization of the Match Flow

For any game, linking the data obtained from the field measurement equipment, the LM-BP neural network we established can update the serve ability x_1 and return ability x_2 in real time. Next, we calculate the momentum value by normalizing x_1 and x_2 . We import the distribution pattern of the indicator data into the Monte Carlo algorithm to randomly generate the momentum value of the two players in the first game, and then use the LM-BP neural network to analyze the change of the momentum of the two players in each of the following games and visualize the flow of the match in the form of a net-chain diagram.

4.4.1 Normality Test

The P-P plot represents the corresponding sample data as scattered points in a right-angled coordinate system, using the cumulative frequency of the samples as the

horizontal coordinate and the cumulative probability (calculated according to a normal distribution) as the vertical coordinate. If the measured sample data follow a normal distribution, the sample points should be distributed around the diagonal of the first quadrant.

The Monte Carlo method can generate a set of random numbers based on the data distribution and parameters of each variable to solve mathematical prediction problems^[11]. Therefore, before using Monte Carlo method to visualize the competition process, we need to test the distribution of the data. We tested the normality of the data of serve ability x_1 and return ability x_2 (the data of x_1 and x_2 were converted from the data in the appendix), and the generated P-P plots are shown below. By analyzing the plot, it can be seen that the distance between these two sets of sample data and the $y = x$ straight line is small, and they are distributed diagonally around the first quadrant. Therefore, it can be judged that the two sets of sample data, the serve ability x_1 and the return ability x_2 , obey a normal distribution.

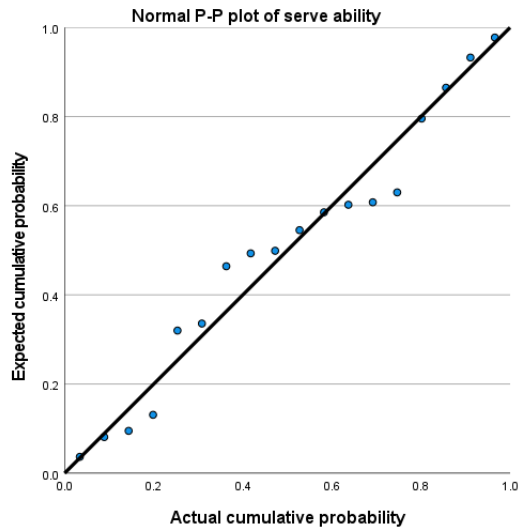


Figure 12 Normal P-P plot of serve ability

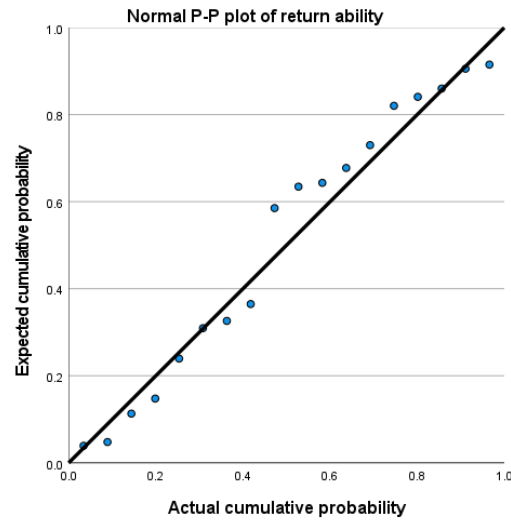


Figure 13 Normal P-P plot of return ability

4.4.2 Visualization Implementation of Match Process

Based on the normality of the samples of serve ability x_1 and return ability x_2 , we use the Monte Carlo method to randomly generate a set of x_1 and x_2 value with normal distribution law. Then normalization is performed to calculate the momentum of the first game of both sides, which is used as the starting point for the iteration of the LM-BP neural network. Through several iterations, we arrive at the momentum of both sides in the next few games, thus grasping the flow of the game, which is visualized in the form of a network chain diagram, as shown in Figure 14 (the numbers on the chain represent probabilities).

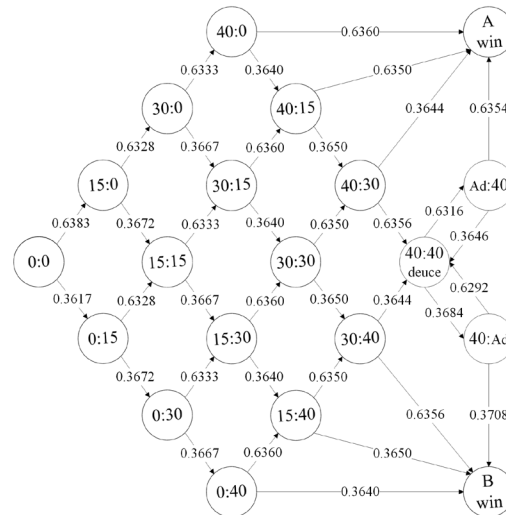


Figure 14 Visual network chain diagram of the match process
(Player A's initial momentum: 0.6383; Player B's initial momentum: 0.3617)

4.5 Model Application 3: The Role of Momentum

We calculated the Spearman correlation coefficient of the score difference of each game between the players and the momentum for the correlation test, and the results are shown in the table below. As can be seen from the table, the correlation coefficient = 0.772 and the significance test $p < 0.001$, which indicates a strong correlation between the two. Therefore, we conclude that the change in momentum can be used to infer the change in score difference and thus make an analysis of the race. As a result, we have overturned the coach's suspicion in Problem 2, and we hold the view that the fluctuation of player performance is related to the winning streak.

Table 3 Correlation test results

Spearman	Scoring margin	Scoring margin		Size of momentum	
		correlation coefficient	1.000	.772**	
		Significance (two-tailed)	.	.000	
		N	150	150	
	Size of momentum	correlation coefficient	.772**	1.000	
		Significance (two-tailed)	.000	.	
		N	150	150	

**. At the 0.01 level(two-tailed),the correlation was significant

5. Model 2: SVM-Based Prediction Model for Match Fluctuations

In this section, we first identify the determinants of match fluctuation, and then establish a classification issue based on the data of 5 matches, and use SVM model to predict match fluctuation by machine learning on historical data. Next, we analyze which factors are highly correlated with match fluctuation by MRMR algorithm. After

that, we predict a match using the established SVM model, calculate its accuracy, and look for reasons for poorly predicted aspects. Finally, we make a qualitative evaluation of the generalization ability of the model.

5.1 Determinants of Match Fluctuations

In the previous analysis, we learned that momentum has a strong positive correlation with the score difference between players in each game and plays an important role in the match. Therefore, we reckon that the match fluctuation is closely related to the momentum change. To avoid the error generated when LM-BP neural network analyzes the momentum to be cumulatively amplified in the prediction process, we choose to make a determination of the match fluctuation in terms of the rate of change in score differentials per game.

When a game is over, if the difference in points changes to less than -2, it means that player 1 is in the disadvantage in this game as he has won only one ball or not won a single ball in this game. In this case, the determinant is denoted as -1; by the same token, if the difference in points is greater than 2, it means that player 1 is in the advantage. When the difference in points changes to -2 or 2, we consider it to be at equilibrium and the determinant is denoted as 0. The decision rules are shown in Table 4.

Table 4 Rules for judging match situation

Score difference per game	-2 or 2	>2	<-2
Fluctuations in matches	0 (Balance)	1 (Strength)	-1 (Weakness)

Based on the situation of the previous game and our prediction of the situation of the next game, determine whether the flow of play is about to change from favoring one player to the other, with the occurrence of fluctuations recorded as 1 and the absence of fluctuations recorded as 0.

5.2 Prediction of Match Fluctuations

SVM models are specialized in classification problems, and show many unique advantages in solving nonlinear and high-dimensional pattern recognition problems, which can basically overcome the problems of "dimensionality catastrophe" and "over-learning" encountered in machine learning. In recent years, SVM models are widely used in time series prediction, text recognition and gene classification.

Recalling the previous conclusions, we know that we can determine whether the match fluctuation occurs through specific variables, and the fluctuation is not only related to the player's offensive situation a_1 and defensive situation a_2 , but also related to the player's error situation a_3 and physical condition a_4 , which involves more than a dozen indicators. In this regard, we build an SVM model for machine learning on historical data to predict match fluctuations.

5.2.1 Establishment Process of SVM Model

The standard SVM model can be reduced to a quadratic programming (QP) problem with constraints, whose goal is to find an optimal classification hyperplane

that meets the classification requirements and maximizes the spacing between the two sides of the hyperplane under the premise of guaranteeing classification accuracy. Theoretically, the SVM model can realize the optimal classification of linearly divisible data, and for nonlinearly divisible data, we can use the kernel function to map it to a suitable feature space to change the nonlinearity of the data. Taking two types of data classification as an example, the specific establishment process of SVM model is shown below:

First of all, we need to give the training sample set $(\mathbf{x}_i, y_i), i = 1, 2, \dots, l, \mathbf{x} \in \mathbf{R}^n, y \in \{\pm 1\}$, and the hyperplane is denoted as $\boldsymbol{\omega} \cdot \mathbf{x} + b = 0$. In order for the classification hyperplane to correctly classify all data with non-zero intervals, the following constraints are to be satisfied:

$$y_i(\boldsymbol{\omega} \cdot \mathbf{x}_i + b) \geq 1, i = 1, 2, \dots, l \quad (8)$$

Next, the size of the classification interval can be obtained as $\frac{2}{\|\boldsymbol{\omega}\|}$ according to the geometric relation. Therefore, the problem of finding the optimal classification hyperplane is transformed into a nonlinear programming problem:

$$\begin{aligned} \min f(\boldsymbol{\omega}) &= \min \frac{1}{2} \|\boldsymbol{\omega}\|^2 = \min \frac{1}{2} (\boldsymbol{\omega}^T \cdot \boldsymbol{\omega}) \\ \text{s.t. } y_i[(\boldsymbol{\omega} \cdot \mathbf{x}_i) + b] &\geq 1, i = 1, 2, \dots, l \end{aligned} \quad (9)$$

For this nonlinear planning problem, we introduce the Lagrange function:

$$L(\boldsymbol{\omega}, b, \mathbf{a}) = \frac{1}{2} \times \|\boldsymbol{\omega}\|^2 - \sum_{i=1}^l a_i [y_i(\boldsymbol{\omega} \cdot \mathbf{x}_i) + b - 1] \quad (10)$$

◆ Where a_i is a Lagrange object

The solution of a nonlinear planning problem is determined by the saddle points of the Lagrange function, i.e., the partial derivatives to $\boldsymbol{\omega}$ and b are satisfied to be zero at the saddle points, which transforms the problem equivalently into the corresponding pairwise problem:

$$\max Q(\mathbf{a}) = \sum_{j=1}^l a_j - \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l a_i \cdot a_j \cdot y_i \cdot y_j \cdot (\mathbf{x}_i \cdot \mathbf{x}_j) \quad (11)$$

$$\text{s.t. } \sum_{j=1}^l a_j \cdot y_j = 0, j = 1, 2, \dots, l, a_j \geq 0, j = 1, 2, \dots, l$$

$$\Rightarrow \text{The optimal solution } \mathbf{a}^* = (a_1^*, a_2^*, \dots, a_l^*)^T$$

Based on the optimal solution $\mathbf{a}^*$, the most weighted vector $\boldsymbol{\omega}^*$ and the optimal weights b^* are then obtained, respectively:

$$\boldsymbol{\omega}^* = \sum_{j=1}^l a_j^* y_j \mathbf{x}_j, b^* = y_i - \sum_{j=1}^l y_j a_j^* (\mathbf{x}_j \cdot \mathbf{x}_j)$$

◆ Where $j \in \{j | a_j^* > 0\}$

Thus, the optimal classification hyperplane $\boldsymbol{\omega}^* \cdot \mathbf{x} + b^* = 0$, can be obtained, and the optimal classification function is $g(\mathbf{x}) = \text{sgn}(\boldsymbol{\omega}^* \cdot \mathbf{x} + b)$. For the linearly indivisible case, we use the kernel function $f(\mathbf{x})$ to map the input vector $\mathbf{X}$ to a high-

dimensional feature space, i.e.

$$X \rightarrow \phi(x) = (\phi_1(x), \phi_2(x), \dots, \phi_l(x))^T \quad (12)$$

At this point, the optimal classification function $g(x)$ is updated to $g(x) = \text{sgn}(\omega \cdot \phi(x) + b) = \text{sgn}(\sum_{i=1}^l a_i y_i \phi(x_i) \phi(x) + b)$.

5.2.2 Analysis of Training Results of SVM Model

In order to comprehensively consider the impact of offensive situation a_1 and defensive situation a_2 , error situation a_3 , and physical condition a_4 on game fluctuations, we extracted the change rate of scores in the appendix and organized it into a response array Y .

Before training the model, we extracted 198 sets of data from the 5 matches in the Appendix, divided them into training and test sets with a 9:1 ratio using the leave-out method, and mapped the initial training data to a higher-dimensional feature space using a polynomial kernel function, and then trained them. Confusion Matrix is shown in Figure 15 and ROC Curve is shown in Figure 16.

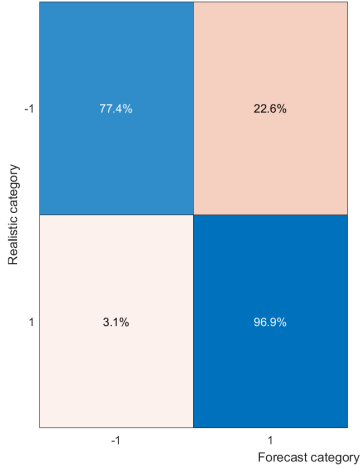


Figure 15 Confusion Matrix Schematic

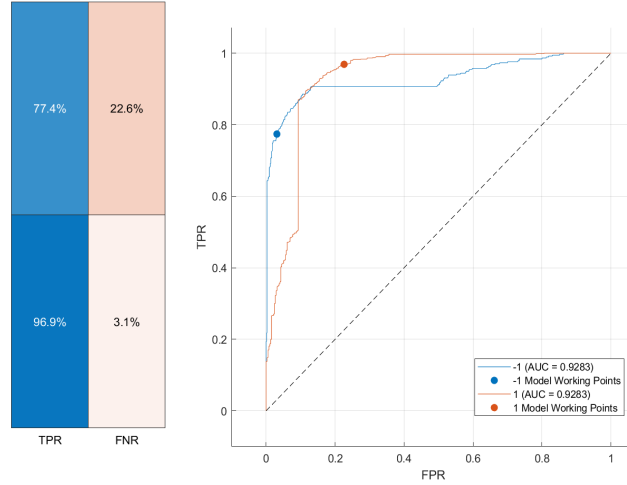


Figure 16 ROC curve schematic

From the confusion matrix, the accuracy of the training model is 90.3% and the recall is 87.15% on average, both at a high level. In the ROC curve, the AUC value of both category 1 and -1 is 0.9283. Since the AUC value characterizes the good or bad classification performance of the model by quantitatively examining the area under the ROC curve, we can conclude that: the match fluctuation prediction model obtained from our training has strong prediction and generalization ability.

In addition, based on the training results, we use the MRMR algorithm to calculate the importance scores of each feature variable by calculating their correlation with the classification or regression target and the redundancy between the feature variables. After normalization, they are sorted by size. The specific results are shown in Figure 17:

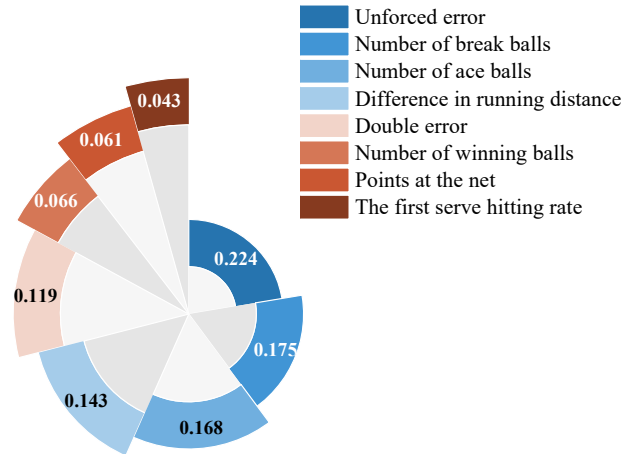


Figure 17 Rose pie chart of importance scores for each indicator

Since the higher the ranking indicates the more important the characteristic variable is in predicting match fluctuations, we can know that the three factors of unforced errors, number of breaks of serve, and number of aces are highly correlated with match fluctuations.

5.3 Advice for Players Dealing with New Opponents

From the above study, we can learn that positive factors such as the first serve hitting rate, ace balls, winning balls, break balls hitting rate, and points at the net, negative factors such as double faults, unforced errors, and physical exertion from running distance will lead to a change in the player's momentum. After a certain gap in players' momentum, the match situation will be skewed from one side to the other. For the purpose of winning a match against a new player, the player should respond as follows:

- ✧ **Enhance personal ability.** Players should hone their serve skills, refine their breaking skills, refine their hitting instincts, and conduct specialized training for aces in their regular training;
- ✧ **Minimize their errors.** Analyze the causes of errors and minimize double faults and unforced errors;
- ✧ **Capitalize on winning opportunities.** Take advantage of winning points and win the match crisply;
- ✧ **Mobilizing the opponent to run the distance.** By adjusting the direction and the strength of the strike, control the landing point of the ball, let the opponent pull back and forth, so as to consume the opponent's stamina, and strive for their own physical advantage;
- ✧ **Observe the opponent and be flexible.** In the first few games of the match, observe and analyze the opponent's playing style and strategy, look for their weaknesses and strengths, so as to adjust the strategy in the next game, and hit the opponent's weaknesses hard while defending well.

5.4 Match Swing Predictions for Wimbledon Men's Final

5.4.1 Accuracy of predicted results

To test the model we developed to apply SVM to predict when the flow of play is about to change from favoring one player to the other, we used the data from the 2023 Wimbledon Men's Single to predict it. After preprocessing the various measurements, we divided them into the set used for training, with the data from the 2023 Wimbledon Men's Final used for testing, and then used the SVM model to make predictions. Using the game's serial number as the horizontal coordinate, we compared the change in the actual score difference with when the flow of play is about to change from favoring one player to the other, and obtained the corresponding prediction image as shown below:

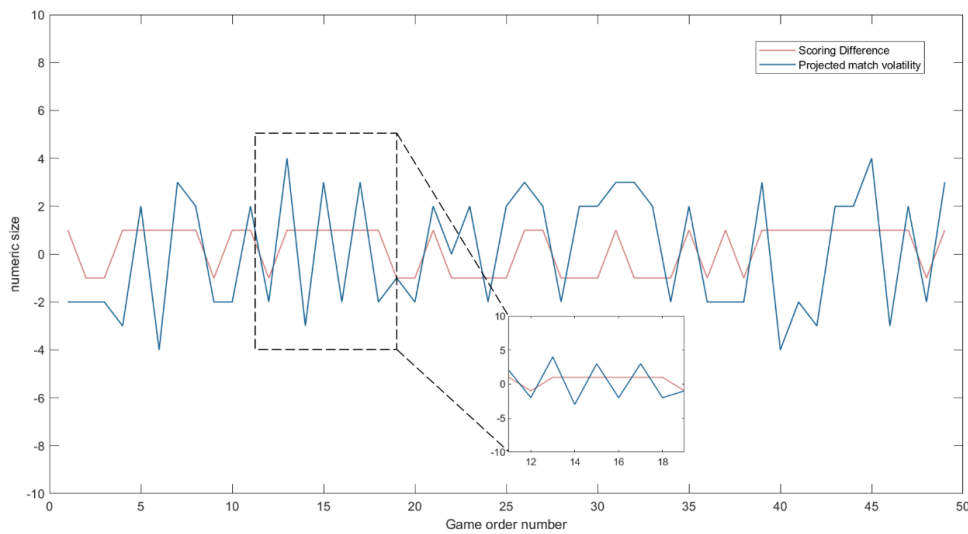


Figure 18 Match Fluctuation Prediction Images

Since when the flow of play is about to change from favoring one player to the other is closely related to the rate of change of the scoring differential, our prediction of match fluctuation should be +1 to show that it is accurate when the trend of the scoring differential is changing. As shown in the graph above, the predicted value was always 1 from game 13 to game 18, i.e., the flow of the game kept changing from favoring one player to the other, and the difference in the actual scores did keep changing. Therefore, we can assume that the prediction of when the flow of play is about to change from favoring one player to the other during this period is accurate. Based on the above principles, we calculate that the model has an accuracy of **83.673%**, which is able to predict match fluctuations to a greater extent.

5.4.2 Error analysis

As observed in the above figure, the part of the model's prediction that is inconsistent with the actual situation is mainly concentrated around the first game of each set, and the difference between the prediction of when the flow of play is about to change from favoring one player to the other around the first game of the first set and the actual situation of the game is due to the fact that we use the historical data to judge the player's momentum, which is at variance with the player's specific performance at

the time of the game. The difference between the prediction of when the flow of play is about to change from favoring one player to the other around the first game of other sets and the actual situation of the actual game is due to the time of rest between sets, the players according to the fluctuation of the game, and adjust accordingly, so the prediction and the actual situation will be different. Therefore, in the future, the model will try to more accurately predict when the flow of play is about to change from favoring one player to the other by adding the covariate of the change in the trend of the players' scores at the beginning of each set in different situations in the past.

5.4.3 Generalization Ability of the Model

For men's singles tennis matches, different tournaments may adopt different systems, such as the four Grand Slam singles matches adopt the best-of-five system, while some other matches adopt the best-of-three system; in addition, the venues of different tournaments are also different, such as the Australian Open and the U.S. Open adopt hard courts, the French Open adopts red clay courts, and Wimbledon adopts grass courts, so the same player's performance and match data on different venues will have a difference. Therefore, the performance and match data of the same player on different venues may vary. Although we can still use the above model to evaluate and predict, the gap between the results and the actual situation may exceed expectations.

For women's tennis, although the selected indicators are almost the same, the size range and weighting of the indicators will change due to gender differences, so the gap between the results of the assessment and prediction using the above model and the actual situation may also exceed the expectation.

As for tournaments of other sports such as table tennis and soccer, the model is no longer applicable due to the change of their selected indicators.

However, no matter for what kind of tournaments, the idea and method of our model are desirable. Our model based on SVM only needs to determine the corresponding features and have sufficient data to make predictions through machine learning. Alternatively, appropriate adjustments or optimizations can be made according to actual needs to achieve higher prediction accuracy.

In summary, our model has a strong generalization ability.

6. Model Evaluation and Analysis

6.1 Sensitivity Analysis

In the process of establishing LM-BP neural networks, we often need to choose the number of hidden layers l ourselves, but there are certain constraints:

$$l = \sqrt{n + m} + a \quad (13)$$

- ◆ Where n is the number of input layer nodes
- ◆ m is the number of output layer nodes, and a is a constant from 0 to 10

To analyze the sensitivity of the LM-BP neural network to the number of hidden layers, based on the same game data, it can be concluded from the above constraints that the number of hidden layers can only be between 4 and 13. By changing the value of the number of hidden layers l for training, the differences in the value of R between

them are compared. The specific results are shown in the following figure:

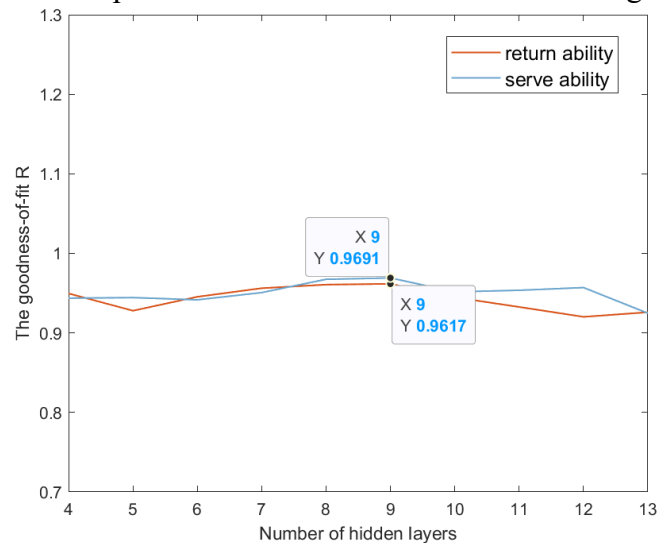


Figure 19 Plot of R-value of training results of LM-BP neural network with different number of hidden layers

From the above figure, it can be seen that whether the training is performed on serve ability X1 or return ability X2, the R values of the training results is greater than 0.9 for any number of implied layers and the maximum magnitude does not exceed 0.035. Therefore, we consider the LM-BP neural network to be more stable, i.e., its results are not subject to changes in the parameters.

6.2 Strength and Weakness

6.2.1 Strength

- When measuring momentum, **we make full use of all the data measured**, taking into account the impact of a variety of factors on a player's momentum, such as: the first serve hitting percentage, rate of break balls, and number of winning balls.
- The Levenberg-Marquardt algorithm is used in Model 1 on top of the classical BP neural networks to **improve the overall training efficiency and reduce the possibility of falling into local optimal solutions**.
- Model 2 In the dichotomous prediction problem of predicting whether a match fluctuation has occurred or not, Model 2 can utilize SVM to **reduce the Hazard of Overfitting** and its kernel trick enables SVM to **handle the nonlinear problem much better**.

6.2.2 Weakness

The evaluation model and prediction model established in this paper **have a large dependence on the data volume of the test data**. Both require machine learning, and without a large amount of test data, the training results are prone to Overfitting or Underfitting.

7 Memorandum

MEMORANDUM

To: Joe Campos, VP of Sales
From: Team # 2416058
Date: February 6, 2024
Subject: The role of momentum



This memo will describe the efficacy of momentum in tennis from three perspectives: the modeling process, the modeling analysis, and the modeling extension, and make relevant suggestions to help tennis players cope with the unknown nature of the game.

I: Modeling process and analysis

(1) Player Momentum Estimation and Game Dynamics Analysis Models

We use the **Levenberg-Marquardt algorithm** based on the Gauss-Newton method with damping to improve the traditional **BP neural network** from the point of view of numerical optimization, and use the improved model to estimate the player's momentum, and the R-value of the model is basically **at a level of more than 0.9**. After that, Spearman's correlation test led us to find that the correlation coefficient between momentum and players' score-per-game **difference was 0.772**, with a **significance test of $p < 0.001$** , which is a strong correlation between the two. This model utilizes the powerful non mapping ability of BP neural network to estimate player momentum through machine learning on a large amount of historical game data, and proves that momentum plays a role in games.

(2) Prediction Model for Match Fluctuations

Match fluctuation prediction is essentially a binary classification problem, we use the linear kernel function to map the data to the appropriate feature space, and then use the **SVM model** to find the optimal classification hyperplane for the feature data, and the accuracy of the final training model is as high as **90.3%**, and the AUC value of positive and negative categories are both **0.9283**. Then, we use the MRMR algorithm to compute the important scores of the feature factors, and we find that the number of unforced errors and the number of aces affect the match fluctuation by **more than 50%**. Next, the **MRMR algorithm** is used to calculate the important scores of each feature factor, and it is found that the degree of influence of unforced errors, number of break balls, and number of ace balls on the match fluctuation is more than 50%, and we utilize the resulting model to predict the fluctuation of a certain match **with an accuracy of 83.673%**.

II: Application of the models

Model 1 is able to use the data distribution to visualize changes in the course of the game based on Markov chain or Monte Carlo method to better understand which player has more momentum at which point in the game.

Model 2 is capable of predicting game fluctuations, and the predicted data can

be used to make flexible changes to your playing tactics during rest periods for match players.

III. Recommendations

According to the calculation of our model above, we can know that momentum can improve the players' performance and affect the winning rate, it is dominating the direction of the game and has a high correlation with the situation on the court; when facing the events affecting the rhythm of the game in tennis, we should keep a good mindset and give full play to our own strength. When facing the events affecting the rhythm of a tennis match, we should keep a good attitude and give full play to our own strengths, because through our model, we can know that the events affecting the rhythm of a tennis match will not completely reverse the momentum.



Sincerely,
Team # 2416058

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